

FairFRL: Fairness-aware Federated Representation Learning for Cross-domain Sequential Recommendation

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Abstract

Cross-domain sequential recommendation is increasingly important in modern Web ecosystems, where user behaviors span multiple independently operated services that maintain strict data isolation for privacy and regulatory compliance. Federated learning offers a practical paradigm for such cross-domain collaboration, but user preferences evolve asynchronously across services, creating a substantial distribution shift. This drift leads to unstable and unequal domain contributions: behaviorally rich domains dominate global updates, while low-resource or volatile domains exert limited influence. Such an imbalance degrades recommendation accuracy and raises fundamental fairness concerns. To address these challenges, we propose FairFRL, a fairness-aware federated representation learning framework designed to mitigate contribution imbalance under dynamic cross-domain drift. FairFRL mitigates contribution imbalance under dynamic cross-domain drift by jointly regulating domain influence during federated aggregation and disentangling domain-shared and domain-exclusive semantics, while preserving data locality. Experiments on real-world Amazon multi-domain datasets show that FairFRL consistently outperforms strong federated and centralized baselines across multiple metrics

and achieves more equitable cross-domain contributions. These results position FairFRL as a principled step toward responsible, fair, and socially aligned Web recommendation systems.

CCS Concepts

• **Information systems** → **Web applications**; • **Computing methodologies** → **Distributed artificial intelligence**.

Keywords

Federated Representation Learning; Cross-domain Sequential Recommendation; Dynamic Bias; Domain-exclusive Modeling; Fairness-aware Aggregation

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1 Introduction

Cross-domain sequential recommendation (CDSR) extends conventional cross-domain recommendation (CDR) by integrating sequential modeling into cross-domain knowledge transfer. Unlike traditional CDR, which treats user behaviors as independent observations, CDSR captures how preferences evolve across domains over time, enabling richer behavioral understanding and more accurate predictions in the target domain [10, 26, 32]. This capability is particularly relevant in modern Web ecosystems, such as

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e-commerce, short-video platforms, and financial services, where user interactions are naturally distributed across heterogeneous service domains. To support such decentralized environments, recent studies have explored privacy-preserving collaborative learning for CDSR, enabling multiple domains to jointly train models without sharing raw data [22, 35]. Meanwhile, the research landscape has evolved from the classic dual-domain setting to more flexible multi-domain and multi-source scenarios [11, 23, 28, 30], reflecting the complexity of real-world user engagement across diverse services.

Although CDSR has achieved promising progress, most existing methods implicitly assume relatively stable or partially overlapping domain distributions [28, 30]. However, real-world user behaviors rarely satisfy these assumptions. Users interact with different services under distinct contexts, and their preferences evolve asynchronously across domains, resulting in heterogeneous and constantly shifting interaction patterns [21, 31]. When domain dynamics diverge, representations learned in one domain may fail to transfer to another, and mismatches in interaction characteristics introduce substantial distribution shifts across domains [33]. In federated or collaborative settings, such a shift directly affects training stability: high-resource domains contribute more frequent and consistent gradients, steering the shared encoder toward their own behavioral patterns, while low-resource domains—whose data distributions fluctuate more dramatically—tend to have suppressed or erratic influence. This imbalance in contribution not only degrades cross-domain generalization but also raises fundamental fairness concerns regarding how heterogeneous domains participate in the shared model. From a Web system perspective, this issue concerns not only predictive performance but also whether heterogeneous domains are equitably represented during collaborative learning.

The contribution imbalance observed in CDSR aligns with fairness challenges in federated learning, where heterogeneous clients influence the global model unequally. Existing fairness-aware federated learning methods, such as minimax optimization [15], loss reweighting [7], dynamic regularization [1], and adaptive aggregation [3, 4, 8], mitigate this issue to some extent. However, these approaches rely on assumptions that do not hold in CDSR. First, they typically assume a shared semantic space, whereas different domains encode distinct behavioral meanings; without separating domain-shared and domain-exclusive factors, dominant domains can overshadow others during joint training. Second, most fairness mechanisms are designed for static or smoothly evolving data, while CDSR exhibits strong temporal contribution drift, causing domain influence to vary as user behaviors evolve. Third, existing approaches regulate aggregation fairness but do not enhance the representational strength of under-resourced or unstable domains, whose semantics remain under-captured even when aggregation weights are adjusted. These limitations indicate the need for a fairness-aware CDSR framework that (i) explicitly disentangles domain-shared and domain-exclusive semantics, (ii) adapts aggregation to time-varying domain contributions, and (iii) strengthens underrepresented domains without introducing additional bias.

In this paper, we present **FairFRL**, a fairness-aware federated representation learning framework for cross-domain recommendation. FairFRL comprises three components. At its core is a *fairness-aware federated knowledge transfer mechanism* that adaptively regulates domain contributions and stabilizes aggregation under dynamic

domain drift. This mechanism is supported by (i) a disentangled representation framework that separates transferable and domain-specific semantics, and (ii) a domain-exclusive latent refinement module that strengthens local behavioral representations via variational modeling and contrastive enhancement.

- We identify **domain contribution imbalance** as a fundamental fairness challenge in federated cross-domain sequential recommendation, arising from heterogeneous and dynamically evolving domain behaviors.
- We propose a **fairness-aware federated aggregation mechanism** that adaptively regulates domain contributions based on their influence on the shared representation and stabilizes training under temporal drift.
- We develop a **disentangled framework with domain-exclusive refinement**, which separates transferable and domain-specific semantics and strengthens local representations without compromising cross-domain fairness.
- Extensive experiments on Amazon cross-domain datasets demonstrate that FairFRL consistently outperforms strong baselines while achieving more balanced and semantically faithful cross-domain knowledge transfer.

2 Related Work

2.1 Cross-Domain Sequential Recommendation

CDSR aims to alleviate data sparsity and cold-start issues by transferring user preference knowledge across domains. Early studies primarily focus on dual-domain alignment. Ma et al. [14] introduce π -Net, a parallel information-sharing network that leverages shared-account interaction sequences for bidirectional preference transfer. They further propose MIFN [13], which combines behavioral and knowledge transfer to capture richer cross-domain dependencies. To enhance representation quality, subsequent works employ contrastive and graph-based strategies. For instance, Cao et al. [2] propose a contrastive infomax framework that jointly models intra- and inter-domain dependencies, while Xu et al. [27] develop MGCL, a multi-view graph contrastive model aligning domain-specific views through item transitions and user preference structures.

With increasing privacy concerns, federated learning has been adopted to enable decentralized CDSR without exposing raw behavioral logs. Ivannikova et al. [5] and Muhammad et al. [16] demonstrate the feasibility of federated collaborative filtering, while Liang et al. [9] propose FedRec++ to handle heterogeneous client distributions. Recent works extend these ideas to cross-domain settings. Yang et al. [29] propose FedGCDR to jointly learn shared and domain-specific graph representations under privacy constraints, and Zheng et al. [36] introduce FedCSR, a dual-contrastive framework that aligns heterogeneous behavioral sequences to mitigate inter-domain semantic gaps.

Despite these advances, existing CDSR and federated CDSR methods largely focus on representation alignment or privacy-preserving optimization, while overlooking domain contribution imbalance under heterogeneous and evolving domain dynamics. In particular, the lack of explicit separation between transferable and domain-exclusive semantics makes shared representations vulnerable to

dominance from high-resource domains and cross-domain distribution shift.

2.2 Fairness in Federated Learning

Fairness has recently become a crucial concern in federated learning, where clients typically vary in data volume, quality, and domain relevance [12, 17]. Early studies primarily adopt static fairness optimization to mitigate performance disparities among clients. Mohri et al. [15] propose agnostic federated learning, a minimax formulation that optimizes the global model for the worst-case client distribution, thereby improving fairness under heterogeneous data conditions. Building on this foundation, Li et al. [7] introduce a fair resource allocation strategy that reweights client losses to minimize performance gaps across participants. Although effective, these approaches rely on fixed weighting schemes and fail to capture the temporal evolution of client contributions, resulting in fairness degradation as data distributions shift over time.

To enhance adaptability, recent works propose dynamic fairness mechanisms that adjust aggregation weights during training. Acar et al. [1] present FedDyn, which introduces a dynamic regularization term to counteract client drift and stabilize global convergence under heterogeneous participation. Cheng et al. [3] design FedGCR, a framework that achieves joint performance and fairness through group customization and adaptive reweighting, allowing clients with distinct data characteristics to receive tailored updates. Similarly, Li et al. [8] propose a client-level dynamic reweighting strategy for long-tailed federated classification, where the contributions of low-frequency or low-resource clients are adaptively amplified during aggregation. Moving beyond reweighting, Gao et al. [4] develop FedCon, a contribution-based aggregation framework that dynamically evaluates client influence to enable scalable and efficient optimization.

Despite these advances, most fairness-aware federated learning methods assume static clients and a homogeneous semantic space, focusing primarily on convergence and efficiency. Such assumptions fail to capture temporal contribution drift and cross-domain semantic heterogeneity, which are inherent to federated cross-domain sequential recommendation.

3 Preliminaries

3.1 Dynamic Federated Optimization

Federated learning aims to collaboratively train a global model from K decentralized clients without sharing raw data. The standard optimization objective is expressed as:

$$\min_{\Theta} \sum_{k=1}^K w_k \mathcal{L}_k(\Theta), \quad (1)$$

where $\mathcal{L}_k(\Theta)$ denotes the local loss of client k , and w_k is its aggregation weight, typically set proportional to the local data volume n_k and normalized such that $\sum_{k=1}^K w_k = 1$. This static weighting scheme implicitly assumes that all clients contribute consistently throughout training, which rarely holds in heterogeneous or evolving environments. To relax this assumption, dynamic federated optimization considers time-dependent aggregation weights that

adapt to clients' temporal behaviors. Let $\Theta^{(r)}$ be the global parameters at communication round r . The conceptual dynamic objective can be written as:

$$\min_{\Theta} \sum_{r=1}^R \sum_{k=1}^K w_k^{(r)} \mathcal{L}_k^{(r)}(\Theta^{(r)}), \quad (2)$$

where $w_k^{(r)}$ is dynamically adjusted at each round based on the client's update behavior or reliability. In practice, this objective is implemented by performing weighted parameter aggregation at each round:

$$\Theta^{(r+1)} = \sum_{k=1}^K w_k^{(r)} \Theta_k^{(r)},$$

which assigns larger aggregation weights to client updates deemed more informative for global optimization.

3.2 Contribution Evaluation and Traceability

During federated learning training, the server receives a sequence of local parameter updates from heterogeneous clients, each following potentially distinct temporal dynamics. To ensure fair and stable aggregation, it is important to *quantify* how each client contributes to global model improvement [20]. Let $\Theta^{(t)}$ denote the global parameters at local step t , and $\Theta_k^{(t)}$ denote the parameters obtained by client k after performing its local update at step t . The local update is defined as:

$$\Delta \Theta_k^{(t)} = \Theta_k^{(t)} - \Theta^{(t)}, \quad (3)$$

which reflects the directional change induced by client k . To measure the influence of this update, a gradient-alignment-based contribution score is used:

$$\gamma_k = \sum_{t \in \mathcal{T}} \left\langle \nabla_{\Theta} \mathcal{L}_{\text{val}}(\Theta^{(t)}), \Delta \Theta_k^{(t)} \right\rangle, \quad (4)$$

where $\mathcal{L}_{\text{val}}$ is the validation loss used to approximate global optimization behavior, and $\langle \cdot, \cdot \rangle$ denotes the inner product. A larger γ_k indicates that client k 's updates are more aligned with reducing global validation loss. Directly computing these gradients at every local step is computationally expensive. Following TracIn-style influence estimation [18], we approximate gradient alignment using parameter differences across checkpoints, enabling efficient client-level contribution tracking throughout the training trajectory. This step-level formulation enables fine-grained analysis of how client updates influence the optimization trajectory over time.

3.3 Problem Definition

We consider cross-domain sequential recommendation in a federated learning environment. There are K domains (clients) $\{\mathcal{D}_k\}_{k=1}^K$, where each domain $\mathcal{D}_k$ holds private user-item interaction sequences $\mathcal{S}_k^u = (s_1^u, \dots, s_{T_u}^u)$, $u \in \mathcal{U}_k$, with s_t^u denoting the item interacted at time t in domain k . Raw interaction data never leave the local domain. The goal is to learn a global sequence prediction model $f_{\Theta} : \mathcal{S}_{k,1:t}^u \mapsto \hat{y}_{u,t+1}$, which predicts the next interaction based on the observed prefix while being collaboratively trained across domains. A key challenge arises from domain heterogeneity: domains differ in data scale, behavioral patterns, and temporal dynamics, causing their local updates to contribute unevenly during federated aggregation. Such imbalance may lead the global model

to overfit high-resource or stable domains while underrepresenting low-resource or fast-changing ones. Therefore, the objective is to learn transferable sequential representations across domains while ensuring fair and balanced domain-level contributions throughout the federated optimization process.

4 Methodology

4.1 Overview

FairFRL is composed of three modules that jointly enable fair and effective cross-domain sequential recommendation under federated settings: (1) disentangled representation learning for separating transferable and domain-specific semantics, (2) fairness-aware federated aggregation for regulating domain contributions during collaborative optimization, and (3) domain-exclusive latent refinement for strengthening local behavioral signals. Figure 1 illustrates the overall architecture.

4.2 Disentangled Representation Learning

We adopt a disentangled representation framework that separates transferable cross-domain preferences from domain-specific behavioral semantics, providing the representational foundation for subsequent federated aggregation.

4.2.1 Latent Space Decomposition. In cross-domain recommendation scenarios, user–item interactions within each domain contain two complementary types of information: transferable preference signals and domain-exclusive behavioral patterns. To explicitly capture this duality, we follow FedDCSR [30] and decompose each user’s latent representation in the k -th domain into a *domain-shared* component $\mathbf{z}_k^s$ and a *domain-exclusive* component $\mathbf{z}_k^e$.

The domain-shared representation $\mathbf{z}_k^s$ is intended to encode preference patterns that generalize across domains and therefore serves as the primary medium for cross-domain knowledge transfer. In contrast, the domain-exclusive representation $\mathbf{z}_k^e$ captures behavioral characteristics that are unique to each domain and preserves domain-specific semantics that should not be conflated during collaborative learning. This explicit decomposition provides the structural foundation for balancing transferability and domain fidelity in federated cross-domain recommendation.

4.2.2 Variational Sequence Encoder. To obtain latent representations from user interaction sequences, we employ a variational sequence encoder with causal self-attention. Given an embedded interaction sequence $\mathbf{E}_k \in \mathbb{R}^{T \times d}$, the encoder processes the sequence using a stack of masked self-attention blocks. Each block consists of masked multi-head attention, residual connections, LayerNorm, and a feed-forward sublayer, enabling the model to capture both short- and long-range dependencies while respecting the autoregressive constraint.

Through these layers, the encoder produces a sequence-level summary vector $\mathbf{h} \in \mathbb{R}^d$, where the causal mask $\mathbf{M}$ prevents any position from attending to future interactions. Based on $\mathbf{h}$, the parameters of a variational posterior are obtained by linear projections:

$$\boldsymbol{\mu} = \mathbf{W}_\mu \mathbf{h} + \mathbf{b}_\mu, \quad \log \sigma^2 = \mathbf{W}_\sigma \mathbf{h} + \mathbf{b}_\sigma, \quad (5)$$

where $\boldsymbol{\mu}, \sigma \in \mathbb{R}^{d_z}$ define a diagonal Gaussian distribution $q_\phi(\mathbf{z} | \mathcal{S}_k^u)$.

A differentiable latent sample is then obtained using the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu} + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (6)$$

which allows gradients to propagate through the stochastic sampling process. The resulting latent vector $\mathbf{z}$ forms the basis for subsequent disentanglement into shared and exclusive components. The domain-shared and domain-exclusive representations serve distinct roles in FairFRL, enabling fair cross-domain transfer while preserving domain-specific semantics. A detailed discussion is provided in Appendix A.

4.3 Fairness-aware Federated Aggregation

This module implements fairness-aware knowledge transfer across domains by regulating how domain-shared representations are aggregated during federated optimization. Building upon the disentangled representation framework in Section 4.2, we focus on controlling domain influence at the aggregation level to mitigate dominance from high-resource or temporally stable domains while preserving training stability.

4.3.1 Shared Encoder. To facilitate domain-shared knowledge transfer across multiple domains, we employ a shared VAE encoder that maps each user sequence $\mathcal{S}_k^u$ to a latent distribution. Specifically, the encoder outputs a variational posterior $q_{\phi^s}(\mathbf{z}_k^s | \mathcal{S}_k^u)$ parameterized by $(\boldsymbol{\mu}_k^s, \sigma_k^s)$ derived from the encoder representation. The domain-shared latent variable $\mathbf{z}_k^s$ is then sampled using the reparameterization operation defined in Equation (6). This shared encoder provides a unified latent space for capturing transferable preference signals across domains.

4.3.2 Federated Aggregation. Each client trains the shared VAE locally using its domain-shared interactions and uploads the updated encoder parameters $\phi_k^{s,(r)}$ to the server at communication round r . The server aggregates these local updates using the dynamic federated optimization rule in Equation (2), producing the global shared encoder parameters for the next communication round. This iterative process incrementally aligns the shared latent representation space across clients while preserving strict data locality.

4.3.3 Dynamic Contribution Evaluation. During each communication round r , client k performs multiple local update steps indexed by t . Since domain-shared representations participate directly in federated aggregation, it is essential that aggregation weights reflect each domain’s actual contribution to the global model. However, user–item interactions evolve over time, causing the utility of client updates to fluctuate. To account for such dynamic influence, we adopt a TracIn-based contribution estimation [18], which quantifies how local updates align with directions that reduce the global validation loss.

Let $\Delta\phi_k^{s,(t)}$ denote the parameter update produced at local step t within round r . The contribution score of client k in round r is computed as:

$$\text{Contrib}_k^{(r)} = \sum_{t \in \mathcal{T}^{(r)}} \left\langle \nabla_{\phi^s} \mathcal{L}_{\text{val}}(\phi^{s,(t)}), \Delta\phi_k^{s,(t)} \right\rangle. \quad (7)$$

Using the standard approximation $\nabla_{\phi^s} \mathcal{L}_{\text{train},k} \approx -\Delta\phi_k^s / \eta$, this computation can be performed efficiently using parameter differences.

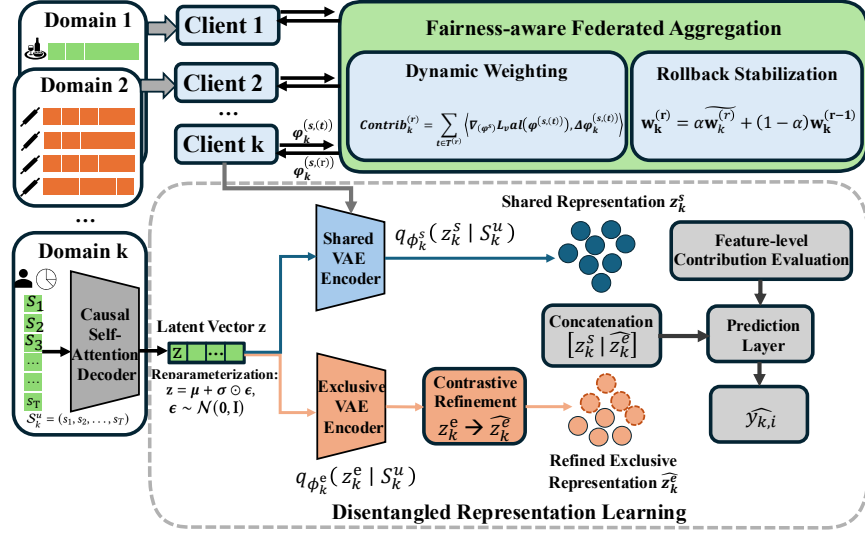


Figure 1: Overall architecture of the proposed FairFRL framework.

Because only domain-shared encoder parameters are uploaded in our framework, this score directly reflects each domain's contribution to shared knowledge transfer.

4.3.4 Rollback Stabilization. Dynamic weighting can be sensitive to noisy or unstable contributions. To ensure stable training, we introduce a rollback-based smoothing mechanism. Let $\tilde{w}_k^{(r)}$ denote the normalized weight derived from $\text{Contrib}(k)$ at round r . The final aggregation weight is updated as:

$$w_k^{(r)} = \alpha \tilde{w}_k^{(r)} + (1 - \alpha) w_k^{(r-1)}, \quad \alpha \in [0, 1]. \quad (8)$$

It is noted that a larger α enables faster adaptation to contribution changes, while a smaller α suppresses oscillation and protects the global model from abrupt degradation.

With dynamic contribution evaluation and rollback stabilization, the federated aggregation process achieves both fairness and robustness, ensuring equitable domain-shared knowledge transfer across evolving user behaviors.

4.4 Domain-exclusive Latent Refinement

This module strengthens domain-specific representations locally to compensate for sparsity and noise in domain-exclusive interactions, without participating in federated aggregation.

4.4.1 Local VAE Encoder for Domain-exclusive Representation. To strengthen representations that reflect behaviors exclusive to a particular domain, each client maintains an independent VAE encoder dedicated to modeling domain-exclusive user interaction sequences. For a user sequence S_k^u that appears only within domain k , the local encoder defines a variational posterior $q_{\phi_k^e}(z_k^e | S_k^u)$, whose mean and variance parameters are computed from the hidden representation of the local encoder. The domain-exclusive latent variable z_k^e is sampled using the reparameterization operation in Equation (6).

In contrast to the shared encoder, the domain-exclusive encoder operates solely within each domain and does not participate in any federated aggregation or cross-domain parameter sharing. This

design enables z_k^e to encode behavioral patterns that belong exclusively to domain k , providing a dedicated representation of domain-exclusive information independently of the shared representation space.

4.4.2 Contrastive-based Latent Refinement. Following FedDCSR [30], we first perform a lightweight intra-domain contrastive enhancement to improve the robustness of the domain-exclusive latent representations. For each domain-exclusive interaction sequence, an augmented view is generated by randomly shuffling the item order (as domain-exclusive interactions are often sparse and heterogeneous, modeling precise temporal order may amplify noise rather than useful semantic signals). Passing both the original and augmented sequences through the exclusive VAE encoder produces a latent pair $(z_k^e, \tilde{z}_k^e)$, which is used to refine the exclusive representation under sparse supervision. The contrastive refinement yields the enhanced domain-exclusive latent $\tilde{z}_k^e$.

4.5 Prediction Layer and Optimization

4.5.1 Prediction Layer. Given the disentangled user representations, the final recommendation score is computed by combining the domain-shared and domain-exclusive latent factors. We first concatenate the two representations:

$$z_k^u = [z_k^s \parallel z_k^e]. \quad (9)$$

The resulting fused representation is then fed into a lightweight prediction layer. Following standard practice in sequential recommendation, we adopt a linear scoring function:

$$\hat{y}_{k,i} = \mathbf{W} z_k^u + b, \quad (10)$$

where $\hat{y}_{k,i}$ denotes the predicted preference score of user u in domain k for item i , and $(\mathbf{W}, b)$ are trainable parameters.

Although simple, this formulation enables the model to leverage both transferable (domain-shared) and domain-exclusive behavioral signals in the final prediction, without introducing additional

modeling bias that may obscure the effects of disentangled and federated representation learning.

4.5.2 Optimization Objective. Our training objective integrates the losses from the shared VAE, the exclusive VAE, the disentanglement constraint, and the contrastive refinement of exclusive latents. The overall loss $\mathcal{L}$ can be formulated as follows:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{VAE}}^s + \gamma \mathcal{L}_{\text{VAE}}^e + \beta \mathcal{L}_{\text{sim}} + \lambda \mathcal{L}_{\text{con}}. \quad (11)$$

Here, $\mathcal{L}_{\text{VAE}}^s$ and $\mathcal{L}_{\text{VAE}}^e$ denote the variational autoencoding losses of the shared and exclusive VAE branches, each composed of a reconstruction term and a KL divergence regularizer. The disentanglement loss $\mathcal{L}_{\text{sim}}$ encourages the separation of domain-shared and domain-exclusive factors, while the contrastive loss $\mathcal{L}_{\text{con}}$ refines the exclusive latent representation via contrastive learning-based augmentation. The details of the full training objective are introduced in Appendix B.

5 Experiments

5.1 Experimental Setup

5.1.1 Datasets. We evaluate FairFRL on three cross-domain sequential recommendation benchmarks derived from the Amazon Product dataset: *Food–Kitchen–Clothing–Beauty (FKCB)*, *Movie–Book–Game (MBG)*, and *Sport–Garden–Home (SGH)*. Each benchmark comprises multiple product categories treated as distinct domains with partially overlapping users, and each domain acts as an independent federated client that keeps its data local. These benchmarks exhibit pronounced domain resource imbalance, making them well-suited for evaluating fairness-aware federated cross-domain recommendation. Details of dataset preprocessing and statistics are provided in Appendix C.

5.1.2 Task Setting. We study the task of **cross-domain sequential recommendation** in a federated learning setting. A federation consists of K clients coordinated by a central server, where each client represents an independent domain (e.g., Food, Movie, or Book) with its own user–item interactions $\mathcal{D}_k = \{\mathcal{U}_k, \mathcal{S}_k^u\}_{u \in \mathcal{U}_k}$. Some users appear in multiple domains, forming transferable behavioral patterns, while others are domain-exclusive. The goal is to collaboratively learn a global model that enhances recommendation performance across all domains by sharing domain-shared representations $\mathbf{z}_k^s$ while keeping domain-exclusive representations $\mathbf{z}_k^e$ and raw data strictly local.

5.1.3 Evaluation Metrics. We evaluate the proposed model from two aspects: recommendation performance and fairness. For recommendation accuracy, we adopt three standard top- N metrics: **MRR**, **HR@10**, and **NDCG@10**. Mean Reciprocal Rank (MRR) measures the average reciprocal rank of the ground-truth item in the predicted list, emphasizing the accuracy of top-ranked results. Hit Rate (HR@10) calculates the proportion of cases where the true next item appears within the top 10 recommendations, while Normalized Discounted Cumulative Gain (NDCG@10) further accounts for the position of the correct item, assigning higher weights to higher ranks. Higher values of all three indicate better recommendation performance. To assess fairness in federated cross-domain learning, we adopt several complementary metrics.

Jain’s Index, **Coefficient of Variation (CV)**, and **Entropy** measure performance balance across domains, where higher Jain’s and Entropy or lower CV indicate better fairness.

5.1.4 Baselines. We compare FairFRL with strong sequential recommendation models adapted to the federated setting, including attention-based architectures [6], variational sequential recommenders [24, 30, 34], and contrastive learning frameworks [19, 25]. Their federated variants are obtained by applying FedAvg to the original models. Baseline descriptions and implementation details are provided in Appendix E and Appendix F, respectively.

5.2 Recommendation Performance

FairFRL demonstrates superior performance across all datasets. As reported in Table 1, FairFRL achieves the highest scores on the FKCB dataset for all evaluation metrics (MRR, HR@10, and NDCG@10). In the *Food* domain, FairFRL yields improvements of **4.42% MRR**, **2.35% HR@10**, and **4.01% NDCG@10** over the strongest baseline. The results in the *Beauty* domain show **4.14% MRR**, **4.95% HR@10**, and **4.50% NDCG@10** increases, illustrating that dynamic aggregation can effectively leverage informative client updates under heterogeneous interaction patterns.

A consistent trend is also observed on the MBG dataset in Table 2. FairFRL achieves domain-wise improvements ranging from **3.11% to 4.72%** across metrics. These results indicate that weighting clients solely by data volume is insufficient when domain distributions differ, whereas aligning aggregation weights with actual contribution yields a more reliable global model.

On the SGH dataset, shown in Table 3, FairFRL maintains the highest performance despite the smaller variance between models. The improvements fall within the range of **0.83%–1.24% MRR** and **0.94%–1.43% NDCG@10**, confirming that FairFRL preserves effectiveness in settings where latent space representations are already compact and well-behaved.

Across all datasets, FairFRL also achieves the best training-size-weighted average (WA), recording gains of **3.25%** on FKCB, **2.47%** on MBG, and **1.12%** on SGH. These results show that contribution-aware aggregation transforms fairness into a measurable performance signal, enabling FairFRL to establish a new state of the art in federated cross-domain sequential recommendation.

5.3 Fairness Analysis

Across all datasets, FairFRL exhibits consistent and robust fairness behavior under varying degrees of domain heterogeneity. The most pronounced improvements are observed on FKCB, which presents the strongest domain imbalance. On this dataset, FairFRL substantially improves Jain fairness for both MRR and HR@10 while notably reducing CV, indicating more balanced and stable cross-domain performance. These gains arise because severe domain imbalance in FKCB amplifies dominant-domain bias, a setting where FairFRL’s dynamic contribution regulation is most effective.

On MBG, where domain distributions are comparatively homogeneous, fairness metrics for all models approach saturation (Jain > 94%). In this regime, FairFRL achieves marginal improvements in HR@10 Jain fairness but slightly underperforms static baselines such as FedDuoRec on MRR Jain and CV. This behavior reflects

Table 1: Cross-domain recommendation performance on the FKCB dataset. All results are reported as mean $\pm$ std (%) over three folds. “WA” denotes the training-size-weighted average over the four domains.

Model	Food			Kitchen			Clothing			Beauty			WA		
	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10
FedCL4SRec	29.11 $\pm$ 0.21	45.81 $\pm$ 0.24	32.63 $\pm$ 0.25	5.78 $\pm$ 0.14	10.11 $\pm$ 0.40	6.11 $\pm$ 0.17	1.87 $\pm$ 0.10	2.81 $\pm$ 0.27	1.75 $\pm$ 0.12	17.78 $\pm$ 0.15	29.51 $\pm$ 0.15	20.02 $\pm$ 0.16	11.44 $\pm$ 0.08	18.67 $\pm$ 0.19	12.61 $\pm$ 0.09
FedContrastVAE	31.42 $\pm$ 0.46	50.14 $\pm$ 0.30	35.47 $\pm$ 0.32	7.66 $\pm$ 0.30	13.87 $\pm$ 0.77	8.38 $\pm$ 0.43	1.93 $\pm$ 0.11	3.02 $\pm$ 0.25	1.80 $\pm$ 0.15	20.16 $\pm$ 0.53	33.13 $\pm$ 0.10	22.72 $\pm$ 0.46	13.09 $\pm$ 0.18	21.74 $\pm$ 0.33	14.58 $\pm$ 0.21
FedDuoRec	29.35 $\pm$ 0.18	46.24 $\pm$ 0.16	32.94 $\pm$ 0.16	5.80 $\pm$ 0.41	9.85 $\pm$ 0.88	6.06 $\pm$ 0.54	1.79 $\pm$ 0.08	2.73 $\pm$ 0.18	1.65 $\pm$ 0.08	18.06 $\pm$ 0.16	29.31 $\pm$ 0.24	20.18 $\pm$ 0.14	11.52 $\pm$ 0.18	18.59 $\pm$ 0.37	12.65 $\pm$ 0.23
FedSASRec	29.88 $\pm$ 0.10	46.26 $\pm$ 0.22	33.36 $\pm$ 0.13	5.75 $\pm$ 0.19	10.02 $\pm$ 0.31	6.07 $\pm$ 0.21	1.71 $\pm$ 0.20	2.73 $\pm$ 0.51	1.58 $\pm$ 0.22	18.34 $\pm$ 0.24	29.63 $\pm$ 0.48	20.51 $\pm$ 0.23	11.64 $\pm$ 0.10	18.72 $\pm$ 0.19	12.78 $\pm$ 0.11
FedVSAN	30.17 $\pm$ 0.38	49.07 $\pm$ 0.68	34.27 $\pm$ 0.41	8.34 $\pm$ 0.28	16.71 $\pm$ 0.58	9.51 $\pm$ 0.33	1.89 $\pm$ 0.27	2.93 $\pm$ 0.52	1.74 $\pm$ 0.30	18.71 $\pm$ 0.77	32.06 $\pm$ 0.76	21.34 $\pm$ 0.65	12.88 $\pm$ 0.20	22.52 $\pm$ 0.33	14.57 $\pm$ 0.21
FairFRL	35.84$\pm$0.30	52.49$\pm$0.26	39.48$\pm$0.31	11.56$\pm$0.51	21.76$\pm$0.79	13.30$\pm$0.58	2.06$\pm$0.41	3.54$\pm$0.84	2.04$\pm$0.51	24.30$\pm$0.23	38.08$\pm$0.25	27.22$\pm$0.22	16.32$\pm$0.24	26.48$\pm$0.38	18.24$\pm$0.28

Table 2: Cross-domain recommendation performance on the MBG dataset.

Model	Movies			Books			Games			WA		
	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10
FedCL4SRec	10.48 $\pm$ 0.22	18.40 $\pm$ 0.30	11.47 $\pm$ 0.19	8.01 $\pm$ 0.06	13.01 $\pm$ 0.17	8.46 $\pm$ 0.10	8.47 $\pm$ 0.15	11.14 $\pm$ 0.08	8.69 $\pm$ 0.09	9.31 $\pm$ 0.10	14.80 $\pm$ 0.16	9.97 $\pm$ 0.12
FedContrastVAE	12.25 $\pm$ 0.03	21.54 $\pm$ 0.12	13.54 $\pm$ 0.04	9.53 $\pm$ 0.12	16.11 $\pm$ 0.20	10.30 $\pm$ 0.15	8.90 $\pm$ 0.19	11.91 $\pm$ 0.18	9.16 $\pm$ 0.13	10.74 $\pm$ 0.05	17.96 $\pm$ 0.14	11.86 $\pm$ 0.07
FedDuoRec	10.33 $\pm$ 0.13	18.03 $\pm$ 0.18	11.26 $\pm$ 0.11	8.10 $\pm$ 0.18	13.19 $\pm$ 0.06	8.57 $\pm$ 0.12	8.17 $\pm$ 0.21	10.78 $\pm$ 0.25	8.38 $\pm$ 0.16	9.38 $\pm$ 0.11	14.86 $\pm$ 0.10	9.94 $\pm$ 0.10
FedSASRec	10.33 $\pm$ 0.26	17.98 $\pm$ 0.55	11.25 $\pm$ 0.30	7.82 $\pm$ 0.29	12.84 $\pm$ 0.48	8.27 $\pm$ 0.32	8.73 $\pm$ 0.17	10.94 $\pm$ 0.27	8.83 $\pm$ 0.21	9.28 $\pm$ 0.13	14.51 $\pm$ 0.31	9.84 $\pm$ 0.16
FedVSAN	7.44 $\pm$ 1.74	16.80 $\pm$ 4.55	8.37 $\pm$ 2.15	7.13 $\pm$ 1.18	12.98 $\pm$ 3.36	7.42 $\pm$ 1.73	2.64 $\pm$ 0.08	5.43 $\pm$ 0.16	2.79 $\pm$ 0.10	6.99 $\pm$ 1.27	13.86 $\pm$ 3.66	7.63 $\pm$ 1.59
FairFRL	16.33$\pm$0.10	28.93$\pm$0.21	18.31$\pm$0.10	13.42$\pm$0.14	23.63$\pm$0.18	14.86$\pm$0.19	11.19$\pm$0.18	16.12$\pm$0.21	12.06$\pm$0.13	14.92$\pm$0.07	25.23$\pm$0.14	16.07$\pm$0.07

Table 3: Cross-domain recommendation performance on the SGH dataset

Model	Sports			Garden			Home			WA		
	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10
FedCL4SRec	5.25 $\pm$ 0.04	6.23 $\pm$ 0.07	5.12 $\pm$ 0.04	5.29 $\pm$ 0.05	5.97 $\pm$ 0.11	5.10 $\pm$ 0.04	4.69 $\pm$ 0.06	5.56 $\pm$ 0.13	4.53 $\pm$ 0.07	5.04 $\pm$ 0.03	5.95 $\pm$ 0.06	4.90 $\pm$ 0.03
FedContrastVAE	5.43 $\pm$ 0.03	6.75 $\pm$ 0.08	5.35 $\pm$ 0.03	5.53 $\pm$ 0.05	6.67 $\pm$ 0.00	5.43 $\pm$ 0.05	4.95 $\pm$ 0.06	6.16 $\pm$ 0.07	4.85 $\pm$ 0.09	5.26 $\pm$ 0.03	6.52 $\pm$ 0.05	5.17 $\pm$ 0.04
FedDuoRec	5.31 $\pm$ 0.06	6.17 $\pm$ 0.23	5.14 $\pm$ 0.10	5.30 $\pm$ 0.14	6.19 $\pm$ 0.10	5.17 $\pm$ 0.13	4.70 $\pm$ 0.04	5.51 $\pm$ 0.07	4.52 $\pm$ 0.05	5.08 $\pm$ 0.04	5.92 $\pm$ 0.12	4.91 $\pm$ 0.06
FedSASRec	5.27 $\pm$ 0.04	6.24 $\pm$ 0.06	5.13 $\pm$ 0.05	5.32 $\pm$ 0.12	6.13 $\pm$ 0.24	5.15 $\pm$ 0.16	4.72 $\pm$ 0.09	5.58 $\pm$ 0.09	4.55 $\pm$ 0.07	5.07 $\pm$ 0.04	5.98 $\pm$ 0.05	4.91 $\pm$ 0.04
FedVSAN	1.53 $\pm$ 0.04	2.62 $\pm$ 0.11	1.34 $\pm$ 0.05	1.53 $\pm$ 0.04	2.40 $\pm$ 0.11	1.32 $\pm$ 0.05	1.92 $\pm$ 0.08	3.69 $\pm$ 0.06	1.86 $\pm$ 0.08	1.68 $\pm$ 0.04	3.00 $\pm$ 0.06	1.53 $\pm$ 0.04
FairFRL	6.26$\pm$0.10	8.45$\pm$0.11	6.29$\pm$0.08	6.77$\pm$0.06	9.09$\pm$0.13	6.86$\pm$0.07	5.94$\pm$0.05	7.64$\pm$0.07	5.91$\pm$0.04	6.19$\pm$0.06	8.21$\pm$0.06	6.21$\pm$0.04

Table 4: Fairness evaluation on the FKCB dataset (3-fold cross-validation). All results are reported as mean $\pm$ std (%). Jain’s index ($\uparrow$) indicates fairness improvement, while CV ($\downarrow$) measures representation stability (lower is better).

Model	MRR		HR@10		NDCG@10	
	Jain $\uparrow$	CV $\downarrow$	Jain $\uparrow$	CV $\downarrow$	Jain $\uparrow$	CV $\downarrow$
FedCL4SRec	61.95 $\pm$ 0.24	78.71 $\pm$ 0.64	63.18 $\pm$ 0.33	76.55 $\pm$ 0.58	60.78 $\pm$ 0.36	80.33 $\pm$ 0.53
FedContrastVAE	64.22 $\pm$ 0.95	74.63 $\pm$ 1.40	65.43 $\pm$ 0.44	72.60 $\pm$ 0.38	63.57 $\pm$ 0.60	75.95 $\pm$ 1.30
FedDuoRec	61.55 $\pm$ 0.62	78.36 $\pm$ 0.97	62.93 $\pm$ 0.57	77.97 $\pm$ 0.89	60.39 $\pm$ 0.77	80.66 $\pm$ 1.09
FedSASRec	61.25 $\pm$ 0.57	79.85 $\pm$ 0.49	62.83 $\pm$ 0.55	77.59 $\pm$ 1.10	60.81 $\pm$ 0.83	81.07 $\pm$ 0.53
FedVSAN	65.94 $\pm$ 0.31	72.98 $\pm$ 0.69	68.16 $\pm$ 0.45	68.67 $\pm$ 0.97	64.83 $\pm$ 0.49	73.32 $\pm$ 0.68
FairFRL	67.94$\pm$0.92	69.70$\pm$1.09	71.05$\pm$1.53	63.32$\pm$1.08	67.87$\pm$1.22	68.14$\pm$2.32

Table 5: Fairness evaluation on the MBG dataset.

Model	MRR		HR@10		NDCG@10	
	Jain $\uparrow$	CV $\downarrow$	Jain $\uparrow$	CV $\downarrow$	Jain $\uparrow$	CV $\downarrow$
FedCL4SRec	98.91 $\pm$ 0.22	11.63 $\pm$ 1.02	95.48 $\pm$ 0.38	21.72 $\pm$ 1.02	98.65$\pm$0.22	14.33 $\pm$ 1.09
FedContrastVAE	98.63 $\pm$ 0.71	14.88 $\pm$ 1.04	94.95 $\pm$ 0.89	23.21 $\pm$ 1.20	97.90 $\pm$ 0.61	17.22 $\pm$ 1.00
FedDuoRec	98.97$\pm$0.27	10.96$\pm$1.22	95.55 $\pm$ 0.45	21.24$\pm$1.01	98.39 $\pm$ 0.46	14.06$\pm$1.02
FedSASRec	98.96 $\pm$ 0.42	11.79 $\pm$ 0.98	95.58 $\pm$ 0.78	21.77 $\pm$ 1.12	98.44 $\pm$ 0.46	14.21 $\pm$ 0.46
FedVSAN	88.09 $\pm$ 4.53	36.54 $\pm$ 6.12	87.58 $\pm$ 4.31	37.96 $\pm$ 6.02	87.24 $\pm$ 4.27	37.59 $\pm$ 6.09
FairFRL	98.89 $\pm$ 0.39	13.72 $\pm$ 1.42	95.91$\pm$0.82	22.67 $\pm$ 0.88	97.81 $\pm$ 0.75	16.73 $\pm$ 1.01

Table 6: Fairness evaluation on the SGH dataset

Model	MRR		HR@10		NDCG@10	
	Jain $\uparrow$	CV $\downarrow$	Jain $\uparrow$	CV $\downarrow$	Jain $\uparrow$	CV $\downarrow$
FedCL4SRec	99.70 $\pm$ 0.54	5.78 $\pm$ 0.61	99.76 $\pm$ 1.26	4.71$\pm$1.69	99.69 $\pm$ 0.91	5.97 $\pm$ 0.80
FedContrastVAE	99.77$\pm$0.20	4.80$\pm$1.11	99.83$\pm$0.82	4.74 $\pm$ 1.16	99.75$\pm$0.52	4.63$\pm$1.00
FedDuoRec	99.66 $\pm$ 0.48	5.83 $\pm$ 0.40	99.66 $\pm$ 1.87	5.62 $\pm$ 1.75	99.60 $\pm$ 1.08	6.31 $\pm$ 1.08
FedSASRec	99.70 $\pm$ 0.93	5.44 $\pm$ 0.82	99.73 $\pm$ 1.24	5.08 $\pm$ 1.43	99.66 $\pm$ 1.02	5.77 $\pm$ 0.87
FedVSAN	98.07 $\pm$ 8.02	10.79 $\pm$ 23.19	96.30 $\pm$ 14.62	19.40 $\pm$ 29.08	97.23 $\pm$ 13.35	16.41 $\pm$ 21.98
FairFRL	99.70 $\pm$ 0.50	5.75 $\pm$ 0.68	99.17 $\pm$ 0.98	7.42 $\pm$ 1.65	99.62 $\pm$ 0.69	6.15 $\pm$ 0.10

the limited room for dynamic fairness mechanisms to provide additional benefit when domain contributions are already stable and balanced.

On SGH, fairness metrics are near their theoretical ceiling (Jain $> 99\%$) and CV values are extremely small. Under this near-uniform setting, FairFRL remains competitive but does not outperform specialized baselines such as FedContrastVAE, as fairness is primarily constrained by the inherent similarity of domain data rather than aggregation strategy.

Overall, these results highlight FairFRL’s *robustness across domain heterogeneity regimes*. While individual baselines may achieve superior fairness under narrowly defined conditions (e.g., static models on balanced domains), FairFRL is the only approach that consistently maintains high Jain fairness and low CV across highly imbalanced, moderately balanced, and fairness-saturated settings. This demonstrates the stability and generalizability of FairFRL’s fairness-aware federated representation learning.

5.4 Ablation Study

To analyze the role of key components in **FairFRL**, we evaluate two ablated variants on the FKCB dataset: (i) disabling dynamic contribution weighting by replacing it with FedAvg, and (ii) disabling the rollback mechanism. The results are reported in Table 7.

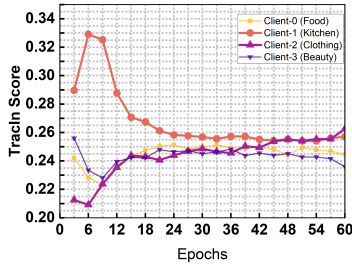
Disabling dynamic weighting leads to noticeable performance degradation on low-resource domains. For example, on the *Clothing* domain, NDCG@10 drops substantially, indicating that static aggregation tends to under-allocate learning capacity to smaller domains. This behavior highlights the role of contribution-aware weighting in alleviating representation dominance during cross-domain training. In contrast, removing the rollback mechanism mainly affects training stability: performance becomes less consistent across domains, with moderate drops observed in domains such as *Kitchen*.

Table 7: Ablation study on the FKCB dataset.

Model Variant	Food			Kitchen			Clothing			Beauty		
	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR	HR@10	NDCG@10
FairFRL-FedAVG	35.35%	52.33%	39.04%	12.03%	24.08%	14.09%	1.90%	3.18%	1.84%	23.36%	36.72%	26.19%
FairFRL w/o rollback	35.17%	52.64%	39.02%	11.78%	21.82%	13.49%	1.93%	3.42%	1.86%	23.84%	38.40%	26.95%
FairFRL (full)	35.44%	52.18%	39.09%	12.30%	22.97%	14.12%	2.65%	4.89%	2.73%	24.66%	38.28%	27.54%

Table 8: Fairness metrics of FairFRL variants (FKCB dataset).

Model Variant	MRR Fairness		HR@10 Fairness		NDCG@10 Fairness	
	Jain ↑	CV ↓	Jain ↑	CV ↓	Jain ↑	CV ↓
FairFRL-FedAVG	0.6788	0.6879	0.7231	0.6188	0.6828	0.6816
FairFRL w/o rollback	0.6788	0.6880	0.7142	0.6326	0.6791	0.6873
FairFRL (full)	0.6963	0.6604	0.7385	0.5951	0.6988	0.6566

**Figure 2: Normalized TracIn scores of the four domains on the FKCB dataset across 60 training rounds.**

This suggests that rollback smoothing helps prevent abrupt shifts in aggregation weights caused by fluctuating domain contributions.

Beyond accuracy, we further examine the impact of each component on cross-domain fairness. As shown in Table 8, the full **FairFRL** model achieves the highest Jain index and the lowest coefficient of variation (CV) across both MRR and HR@10. Replacing dynamic weighting with FedAvg yields fairness scores comparable to standard federated baselines, indicating that without contribution-aware aggregation, large domains dominate global updates. Removing rollback stabilization also degrades fairness by increasing CV, reflecting the propagation of instability across communication rounds.

Overall, the ablation results clarify the complementary roles of FairFRL’s components: dynamic contribution weighting primarily improves domain-level allocation fairness, while rollback stabilization enhances robustness under temporal contribution drift. Together, these mechanisms enable balanced and stable cross-domain knowledge transfer in heterogeneous federated settings.

5.5 Visualization Analysis of TracIn Scores

Figure 2 provides a process-level view of how domain contributions evolve during federated training on the FKCB dataset. In the early training stages, the *Kitchen* domain exhibits a relatively high normalized TracIn score, while *Clothing* contributes noticeably less. This pattern reflects an initial contribution imbalance, where domains with richer or more stable interactions exert a stronger influence on the shared model. As training progresses,

the contribution-aware aggregation mechanism gradually moderates these disparities. After approximately 20 communication rounds, TracIn scores across the four domains converge to a narrow range and remain stable thereafter, indicating a more balanced contribution pattern. This temporal behavior is consistent with the improvements in Jain’s index and the reductions in coefficient of variation (CV) observed on FKCB, suggesting that early-stage dominance is effectively mitigated over the course of training.

Overall, Figure 2 offers an interpretable perspective on FairFRL’s fairness dynamics by illustrating how domain influence is regulated over time. Rather than focusing on peak contributions, FairFRL promotes sustained balance across domains, helping explain the fairness improvements observed in the quantitative evaluation.

6 Limitations and Future Work

FairFRL regulates domain contributions during federated optimization but does not increase data volume of low-resource domains. While domain-exclusive refinement improves representation robustness, sparse domains may remain semantically underrepresented. Moreover, FairFRL targets domain-level fairness, which is natural in federated settings but does not capture finer-grained notions such as user-level or subgroup-level equity. Finally, its benefits diminish when domain distributions are homogeneous or fairness metrics approach saturation. Addressing data scarcity and developing adaptive, fairness-aware augmentation mechanisms under privacy constraints remain important directions for future work.

7 Conclusion

We presented FairFRL, a fairness-aware federated representation learning framework for cross-domain sequential recommendation. By disentangling domain-shared and domain-exclusive behaviors and dynamically regulating aggregation via contribution-aware weighting and rollback stabilization, FairFRL mitigates dominant-domain bias under heterogeneous and evolving domain dynamics. Experiments on multiple Amazon benchmarks demonstrate that FairFRL achieves competitive accuracy while consistently improving domain-level fairness. More broadly, this work highlights domain participation as a first-class fairness concern in decentralized recommender systems and provides an interpretable foundation for responsible and socially aligned federated recommendation.

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A Shared vs. Exclusive Representation Roles

The domain-shared and domain-exclusive representations in FairFRL serve distinct functional roles and are subject to different optimization objectives and information constraints.

The domain-shared representation $\mathbf{z}_k^s$ captures transferable preference patterns that generalize across domains. It participates in federated optimization and is the only latent factor whose encoder

parameters are communicated across clients. Its learning is regulated to ensure balanced cross-domain influence and to mitigate representation bias induced by dominant domains.

In contrast, the domain-exclusive representation $\mathbf{z}_k^e$ encodes domain-specific behavioral characteristics and remains strictly local. It is never shared or aggregated, preserving domain semantics without leakage. Since exclusive interactions are often sparse or noisy, $\mathbf{z}_k^e$ is optimized independently within each domain and re-fined to improve robustness.

This asymmetric design separates responsibilities: shared representations enable fair and transferable cross-domain learning, while exclusive representations preserve domain individuality. Without such disentanglement, dominant domains may bias the shared latent space, leading to negative transfer and unfair representation imbalance.

B Full Training Objective

This appendix summarizes the complete training objective of FairFRL. The overall loss consists of two VAE-based representation losses and two regularization terms, which are jointly optimized during training.

B.1 VAE-based Representation Losses

Shared VAE Loss. For each user sequence $\mathcal{S}_k^u$, the shared encoder parameterizes a diagonal Gaussian posterior:

$$q_{\phi^s}(\mathbf{z}_k^s | \mathcal{S}_k^u) = \mathcal{N}(\mu_k^s, \text{diag}((\sigma_k^s)^2)).$$

The shared VAE loss is defined as:

$$\mathcal{L}_{\text{VAE}}^s = \mathcal{L}_{\text{recon}}^s + \text{kld_weight} \cdot \text{KL}(q_{\phi^s}(\mathbf{z}_k^s | \mathcal{S}_k^u) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})), \quad (12)$$

where the KL divergence has the closed form

$$\text{KL}_s = -\frac{1}{2} \sum_{i=1}^d \left(1 + \log(\sigma_{k,i}^s)^2 - (\mu_{k,i}^s)^2 - (\sigma_{k,i}^s)^2 \right), \quad (13)$$

and $\mathcal{L}_{\text{recon}}^s$ denotes the reconstruction loss.

Exclusive VAE Loss. The domain-exclusive encoder similarly defines a posterior $q_{\phi^e}(\mathbf{z}_k^e | \mathcal{S}_k^u)$. Its VAE loss is given by

$$\mathcal{L}_{\text{VAE}}^e = \mathcal{L}_{\text{recon}}^e + \text{kld_weight} \cdot \text{KL}(q_{\phi^e}(\mathbf{z}_k^e | \mathcal{S}_k^u) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})), \quad (14)$$

with the KL divergence computed analogously to the shared branch.

B.2 Disentanglement and Regularization Losses

Shared-Exclusive Disentanglement. To enforce separation between shared and exclusive latent factors, we adopt a contrastive-style disentanglement loss:

$$\mathcal{L}_{\text{sim}} = \max(0, m - \text{sim}(\mathbf{z}_k^s, \mathbf{z}^g) + \text{sim}(\mathbf{z}_k^e, \mathbf{z}^g)), \quad (15)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and m is a margin. This term encourages $\mathbf{z}_k^s$ to align with the global semantic space $\mathbf{z}^g$ while pushing $\mathbf{z}_k^e$ away.

Intra-domain Contrastive Refinement. Each domain-exclusive sequence is paired with an augmented view $\tilde{\mathcal{S}}_k^u$ obtained by random item shuffling. Passing both sequences through the exclusive encoder yields $(\mathbf{z}_k^e, \tilde{\mathbf{z}}_k^e)$. The contrastive loss is defined as

$$\mathcal{L}_{\text{con}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_k^e, \tilde{\mathbf{z}}_k^e)/\tau)}{\sum_{u' \neq u} \exp(\text{sim}(\mathbf{z}_k^e, \tilde{\mathbf{z}}_{k'}^e)/\tau)}, \quad (16)$$

where τ is a temperature parameter.

B.3 Overall Objective

The complete training objective of FairFRL is

$$\mathcal{L} = \alpha \mathcal{L}_{\text{VAE}}^s + \gamma \mathcal{L}_{\text{VAE}}^e + \beta \mathcal{L}_{\text{sim}} + \lambda \mathcal{L}_{\text{con}}. \quad (17)$$

C Dataset Preprocessing and Statistics

All experiments are conducted on three cross-domain sequential recommendation benchmarks derived from the Amazon Product dataset: Food-Kitchen-Clothing-Beauty (FKCB), Movie-Book-Game (MBG), and Sport-Garden-Home (SGH). Each benchmark consists of multiple product categories, where each category is treated as an independent domain (client) in the federated learning setting.

Following common practice in sequential recommendation, we filter out users and items with fewer than ten interactions to ensure sufficient behavioral continuity. All remaining user-item interactions are chronologically sorted to construct domain-specific interaction sequences. For each user u in domain k , the interaction sequence is denoted as $\mathcal{S}_k^u = (s_1^u, \dots, s_{T_u}^u)$, where T_u is the sequence length. After preprocessing, sequence lengths range from 4 to 16 across all benchmarks.

For each domain, interaction sequences are split into training, validation, and test sets with an 8:1:1 ratio. User identifiers are globally aligned across domains to enable cross-domain preference modeling, while all raw interaction data remain strictly local throughout training. Each domain participates as an independent federated client, and only model parameters are exchanged during federated optimization. Detailed dataset statistics after preprocessing are reported in Table 9.

D Evaluation Metrics

This section provides detailed definitions of all evaluation metrics used in this study, encompassing both recommendation performance and fairness metrics.

D.1 Recommendation Performance Metrics

We adopt three widely used top- N metrics to evaluate sequential recommendation performance:

(1) **Mean Reciprocal Rank (MRR).**

$$\text{MRR} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{1}{\text{rank}(s_{t+1}^u)}, \quad (18)$$

where $\mathcal{U}$ denotes the set of users, and $\text{rank}(s_{t+1}^u)$ is the position of the true next item s_{t+1}^u in the ranked list of predictions for user u .

(2) **Hit Rate (HR@N).**

$$\text{HR@10} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \mathbb{I}(s_{t+1}^u \in \text{Top}_{10}(\hat{\mathbf{y}}_{u,t+1})), \quad (19)$$

where $\hat{\mathbf{y}}_{u,t+1}$ represents the model's predicted ranking scores for the next item of user u , and $\mathbb{I}(\cdot)$ is an indicator function that equals 1 if the ground-truth item appears within the top-10 predictions.

Table 9: Statistics of the Amazon datasets.

Statistic	Food	Kitchen	Clothing	Beauty	Movie	Book	Game	Sport	Garden	Home
Total Users	4658	13382	9240	5902	34792	19419	5588	28139	6852	20784
Total Items	13564	32918	34909	17780	44464	72246	10336	88992	21604	62499
Training Set	4977	11100	5720	4668	57405	63157	6631	51477	10479	37361
Valid Set	1307	2172	818	836	10944	11168	1374	13720	3074	10058
Testing Set	1332	2254	866	855	11654	12149	1444	14214	3113	10421
Average Lenth	8.79	8.60	9.30	10.29	7.97	7.20	6.49	10.65	9.48	10.41

(3) Normalized Discounted Cumulative Gain (NDCG@N).

$$\text{NDCG@10} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{1}{\log_2(1 + \text{rank}(s_{t+1}^u))}, \quad (20)$$

which assigns higher weights to higher-ranked correct predictions. Larger values of MRR, HR@10, and NDCG@10 indicate better recommendation performance.

D.2 Fairness Evaluation Metrics

To measure fairness in federated cross-domain learning, we adopt multiple complementary metrics capturing balance, stability, and representational consistency.

(1) Jain's Index.

$$\text{Jain} = \frac{\left(\sum_{k=1}^K r_k \right)^2}{K \sum_{k=1}^K r_k^2},$$

where r_k denotes the performance (e.g., HR@10) of domain k , and K is the total number of domains. It measures the uniformity of performance distribution across domains; higher values indicate greater fairness.

(2) Coefficient of Variation (CV).

$$\text{CV} = \frac{\sigma(r)}{\mu(r)},$$

where $\sigma(r)$ and $\mu(r)$ are the standard deviation and mean of $\{r_k\}$, respectively. A lower CV implies smaller performance disparity among domains.

E Baseline Models

To provide a comprehensive comparison, we include representative sequential recommendation methods spanning three modeling paradigms. All baselines are trained under the same federated protocol by applying FedAvg for parameter aggregation.

- **Attention-based Sequential Models.** SASRec [6] adopts self-attention blocks to capture long-range patterns in user histories. We extend it to a federated environment and denote the resulting model as **FedSASRec**.
- **Variational Sequential Models.** VSAN [34] introduces stochastic sequence encoding via variational self-attention,

and ContrastVAE [24] integrates variational inference with contrastive regularization for stable latent representations. Their federated counterparts are referred to as **FedVSAN** and **FedContrastVAE**. In addition, **FedDCSR** [30] applies disentangled variational modeling to separate domain-shared and domain-exclusive behaviors in federated settings.

- **Contrastive Learning-based Models.** CL4SRec [25] exploits augmented sequence views for self-supervised contrastive alignment, and DuoRec [19] enhances robustness using dual contrastive objectives. We adopt their federated adaptations, denoted as **FedCL4SRec** and **FedDuoRec**.

F Implementation Details

We implement all experiments using PyTorch 2.7.0 with CUDA support and NumPy 1.26.4 under Python 3.12.6. All experiments are conducted on an NVIDIA RTX4080s GPU with 16 GB of memory. Each client is trained locally with a batch_size of 256 and a maximum sequence length of 16, where shorter interaction sequences are padded, and longer ones are truncated. The model is optimized using the Adam optimizer with an initial learning rate of 0.001 and a weight decay of 0.0005. The global federated training runs for 40 communication rounds, and each client performs 3 local epochs per round. The learning rate decays by a factor of 0.1 every 10 epochs if no improvement is observed on the validation set. Early stopping is applied with a patience of 5 rounds. Following prior works [30, 34], we set the temperature of the contrastive loss to 1.0 and employ dot-product similarity. All reported results are obtained using a 3-fold cross-validation setup to ensure robustness and statistical reliability. Code implementation is available at <https://doi.org/10.5281/zenodo.18315774>.

For verifiable training, the TracIn-based contribution evaluation is executed at every federated training round, enabling round-wise contribution estimation throughout optimization. In contrast, the Influence-Function-based contribution analysis is performed only after training is completed, using the checkpoints saved every 5 rounds to conduct retrospective LiSSA-based inverse-Hessian estimation. The LiSSA procedure uses 10 iterations with a damping coefficient of 0.01 to ensure stable Hessian-vector approximation. All random seeds are fixed to 42 for reproducibility.